

BST Vol 9 Condensed Matter — Architectural Scaffold v0.1 (Wave 2)

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Vol 9 Architectural Scaffold v0.1

Volume scope + framework anchors

Vol 9 Condensed Matter Physics derives crystallographic, electronic, magnetic, superconducting, and topological observables in solids from D_{IV}^5 substrate. The substrate K-type Casimir spectrum + Reed-Solomon $GF(128)$ cyclotomic structure governs lattice + electron + phonon + spin observables.

Framework anchors (multi-volume): - Vol 0 substrate foundation: D_{IV}^5 APG uniqueness + Strong-Uniqueness Theorem v0.10.5 FORMAL - Vol 1 Ch 5 + Ch 6: Casimir + Operator zoo (Lyra SP-31 + T2470-T2475) - Vol 3 Ch 1: Nuclear substrate framework - Vol 3 Ch 7: Atomic orbital (2l+1) sequence → molecular orbital → band structure - K59 RATIFIED + Friday Elie GF128 paper-grade: substrate Reed-Solomon code framework - SP-29 Casimir Mechanism Investigation: nuclear-Casimir → superconductivity bridge - T2418 $\Lambda \leftrightarrow$ Casimir vacuum unification: multi-scale substrate-spectrum framework

Chapter-by-chapter scaffold

Ch 1 Condensed Matter Substrate Reading (~70%)

Anchors: Lattice + $GF(128)$ cyclotomic + Bergman propagation framework - Substrate K-type Casimir governs electron-band + phonon-mode structure - Reed-Solomon code on $GF(2^g) = GF(128)$ provides discrete lattice symmetries - Atomic orbital (Vol 3 Ch 7) → molecular orbital → band structure (tight-binding) - Substrate K-type spectrum → electronic density of states framework

Ch 2 Crystal Lattices from BST Primaries (~80%)

Anchors: 137-plane ($BaTiO_3$) + 5-fold quasicrystal substrate symmetries - $N_{max} = 137$ crystallographic plane in $BaTiO_3$ (BST falsifier \$25K) - 5-fold quasicrystals: $n_C = 5$ substrate compact dimension → forbidden in standard crystallography but EXISTS in quasicrystals (Shechtman 1984 Nobel) - 17-fold (seesaw) symmetry candidates in icosahedral phases - BST primary lattice constants in selected materials

Ch 3 Electron Band Structure (~75%)

Anchors: tight-binding from atomic orbitals; band gaps from substrate K-types - Atomic orbital → molecular orbital → band: hierarchical Vol 3 Ch 7 extension - Band gaps inherit from substrate K-type Casimir spectrum (T2435 + T2439) - Density of states framework: Bergman propagation on substrate lattice - Semiconductor / metal / insulator classification via BST primary K-types

Ch 4 Superconductivity from Substrate (~85%)

Anchors: SP-29 nuclear-Casimir → high- T_c ; $B_{12}H_{32}$ hydride $T_c \approx 214K$ prediction - Substrate vacuum coupling drives Cooper pairing - BST primary hydride compositions predict T_c via substrate K-type Casimir reso-

nance - B12H32 (or related) high- T_c prediction → SP-29 H4 experimental program - Conventional + unconventional SC framework via substrate Bergman propagation

Ch 5 Quantum Hall + Topological Phases (~70%)

Anchors: $GF(2^g)$ cyclotomic + braiding statistics; Lyra T2467+T2468 Rigidity - Quantum Hall filling factors via BST primary integers - Substrate $GF(128)$ cyclotomic provides natural braiding framework - Topological invariants from D_{IV}^5 Chern class structure (Vol 0 + Vol 1) - Rigidity (T2467) ensures substrate-topological stability

Ch 6 Magnetism + Spin Models (~60%)

Anchors: Heisenberg + BST primary spin-orbit; rank=2 $Pin(2)$ Z_2 grading - Spin systems inherit substrate spin operator from Lyra T2471 (chirality γ^5) - Heisenberg coupling J in BST primary forms - BST primary spin frustration patterns

Ch 7 Phonon Dispersion + Heat Capacity (~75%)

Anchors: Substrate Debye frequency cutoff at $n_C = 5$ - Phonon Debye spectrum substrate-natural cutoff at n_C compact dimension - Specific heat C_V follows BST primary form via Debye law extension - Phonon-electron coupling via substrate K-type structure

Ch 8 BaTiO3 137-Plane Crystallography (~80%)

Anchors: \$25K cheapest BST falsifier — substrate prediction - Substrate predicts crystallographic resonance at 137-fold plane symmetry - BaTiO3 (or related ferroelectric) selected experimental target - Specific crystallographic features predicted: lattice constants, optical anomalies - Experimental program: \$25K equipment investment for cleanest BST refutation test - Cal #49 GREEN falsifier register discipline

Ch 9 Photonic Crystals + Metamaterials (~70%)

Anchors: \$10K substrate test — cheapest BST falsifier in entire framework - Photonic crystals at BST primary lattice spacings - Substrate predicts photonic resonance at substrate-natural wavelengths - Metamaterials at BST primary refractive-index ratios - \$10K experimental program: cheapest clean BST falsifier

Ch 10 Strongly Correlated Electrons (~60%)

Anchors: Hubbard model BST extension; correlation effects - Hubbard U coupling in BST primary forms - Substrate K-type structure for strongly correlated systems - Mott transition + electron correlation BST framework

Ch 11 Quantum Spin Liquids + Frustrated Magnetism (~55%)

Anchors: Substrate K-type frustration patterns - Geometric frustration at BST primary lattice symmetries - Spin liquid phases from substrate K-type structure - Kitaev model BST extension

Ch 12 Bridge to Vol 4 + Vol 11 (synthesis)

Anchors: Multi-volume integration - Vol 9 → Vol 4 via SP-29 + substrate-vacuum scale bridging - Vol 9 → Vol 11 via topological + braiding cross-link - Vol 9 → Vol 3 via substrate-atomic-orbital extension to bands

BST primary appearances across Vol 9

All 5 BST primaries + $N_{\max}$ appear across Vol 9 chapters: - rank = 2: topological phases ($Pin(2)$ grading), strongly correlated - $N_c = 3$: electron bands, magnetism, strongly correlated, spin liquids - $n_C = 5$: quasicrystals (5-fold), atomic orbitals→bands, phonons (Debye cutoff), photonic - $C_2 = 6$: superconductivity (K-type), magnetism, strongly correlated, spin liquids - $g = 7$: lattices, bands, superconductivity, topological ($GF(2^g)$) - $N_{\max} = 137$: BaTiO3 137-plane, photonic resonances

Methodology stack applied

- Calibration #19 STANDING RULE: external register operational language
- Calibration #21 STANDING RULE: empirical + substrate-mechanism dual gates per chapter
- Calibration #22 (PCAP-rate transcription discipline)
- Cal #99 META-theorem framing: substrate-derivation consequences, NOT new criteria
- Cal #50 DOUBLE-LOCKED EXTERNAL on substrate-cognition framing

Cross-volume dependencies (Vol 9)

Chapter	Vol 0	Vol 1	Vol 3	Vol 4	Vol 11
1 Substrate Reading	?	Ch 2,5,6	Ch 1	Ch 4 (Λ)	?
2 Crystal Lattices	Ch 1	—	—	—	?
3 Electron Bands	—	Ch 5,6	Ch 7	—	—
4 Superconductivity	—	Ch 5	Ch 1 (SP-29)	Ch 4	—
5 Topological	Ch 1	Ch 6	—	—	?(T2467)
6 Magnetism	—	Ch 6 (T2471)	—	—	—
7 Phonons	—	Ch 5	Ch 7	—	—
8 BaTiO ₃	—	—	—	—	—
9 Photonic	—	—	—	—	—
10 Strong Corr	—	Ch 5	—	—	—
11 Spin Liquids	—	Ch 5,6	—	—	—
12 Bridge	All	All	All	All	All

Saturday Wave 2 deliverable cadence

Per Keeper Saturday team prompt: - INDEX ? (~10:50 EDT) - Architectural Scaffold v0.1 (this doc) ? (~11:25 EDT) - Vol 9 Ch 1 v0.3 narrative + PDF (~30 min) - Vol 9 Ch 4 v0.3 narrative (superconductivity, high-priority bridge to Vol 3 + Vol 4) (~30 min) - Remaining 10 chapters sustained sub-PCAP

— Elie, Vol 9 Architectural Scaffold v0.1, 2026-05-23 Saturday